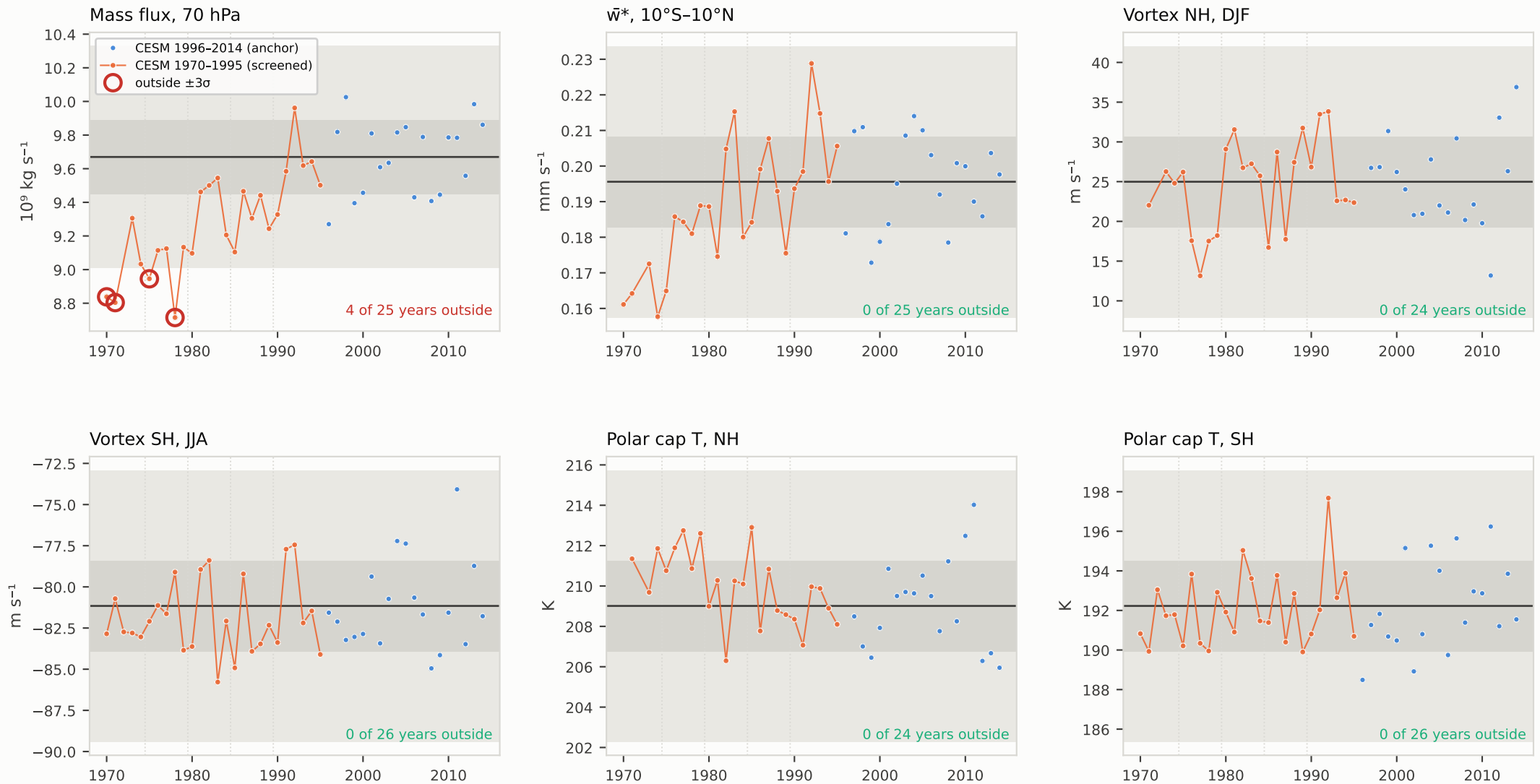


Tier-1 screening, applied to CESM 1970-1995

Every individual year of 1970-1995 (orange) scored against the $\pm 3\sigma$ screening band set on 1996-2014 (blue). Dark grey = $\pm 1\sigma$, where CESM's own years mostly live; light grey = the $\pm 3\sigma$ gate. Red rings mark years the gate rejects. Dotted verticals are the 5-year screening blocks. Only the mass flux fails, and it fails in the 1970s — that is the forced BDC trend (D9), not a broken configuration.



The same thing as screening-run verdicts

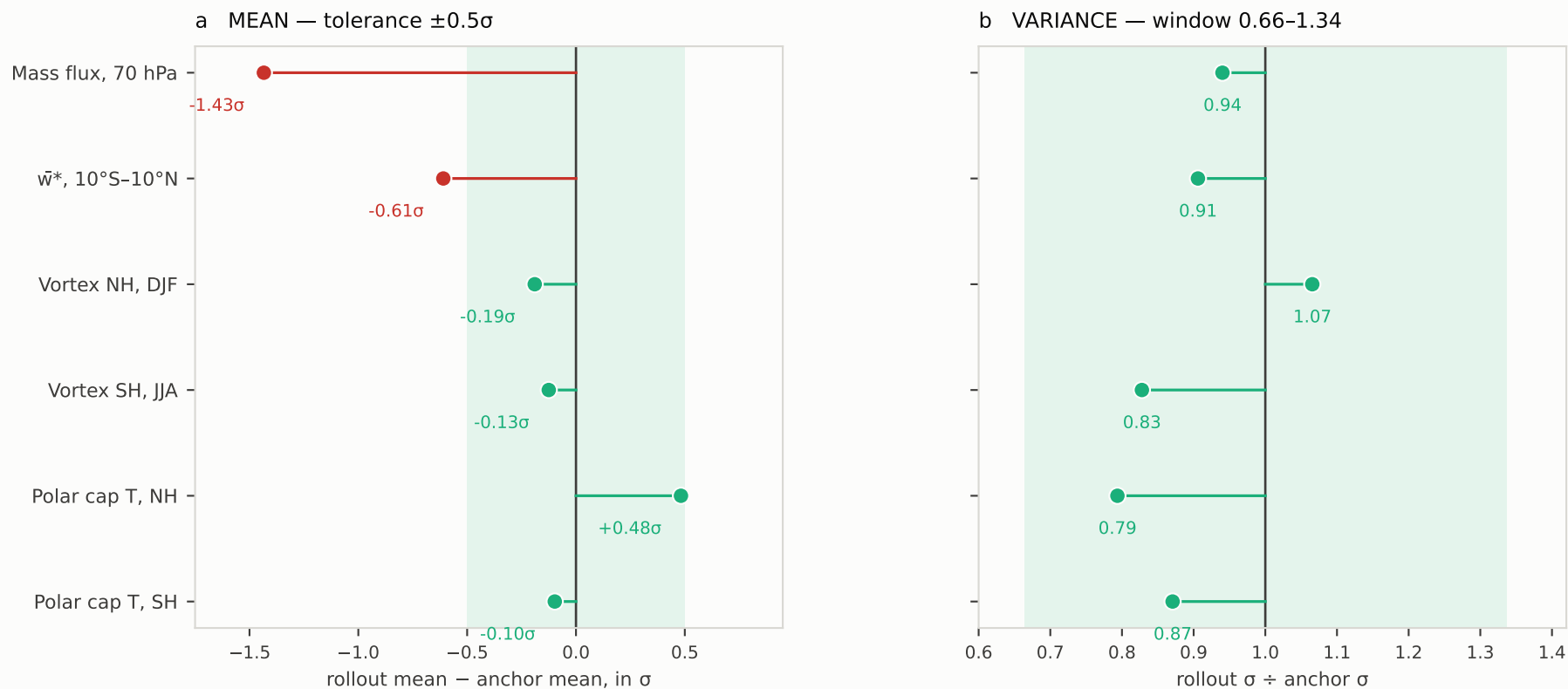
1970–1995 split into five consecutive 5-year blocks, each treated as one screening run against the $\pm 3\sigma$ band set on 1996–2014.

	1970–74	1975–79	1980–84	1985–89	1990–94
Mass flux, 70 hPa	FAIL 70 71	FAIL 75 78	pass	pass	pass
$\bar{w}^*$, 10°S–10°N	pass	pass	pass	pass	pass
Vortex NH, DJF	pass	pass	pass	pass	pass
Vortex SH, JJA	pass	pass	pass	pass	pass
Polar cap T, NH	pass	pass	pass	pass	pass
Polar cap T, SH	pass	pass	pass	pass	pass

2 of 30 block verdicts fail — both mass flux, both before 1980, 4 failing year-checks in total. That is the forced BDC trend (D9), not a broken configuration: the 1970–1995 mass flux sits 1.77σ below the anchor. The other five diagnostics never flag in 26 years of out-of-sample CESM, which is the behaviour a screening gate needs — it does not fire on a healthy model.

Tier 2 — CESM 1970-1994 scored against the 1980-2014 anchor

The two things tier 2 tests. Left: the rollout MEAN against the $\pm 0.5\sigma$ tolerance. Right: the rollout σ against the 35-vs-35-year acceptance window. σ is detrended within each window where the trend is significant, as in 07_period_split. The windows overlap by 15 years (1980-1994), so this is a consistency check, not an independent out-of-sample test — and 1980-2014 spans the 1995/96 restart between two separate CESM runs.



4 of 6 diagnostics pass the mean test and 6 of 6 pass the variance test. The two failures are the upwelling pair, and they fail low— mass flux at -1.43σ and $\bar{w}^*$ at -0.61σ against a $\pm 0.50\sigma$ tolerance. That is the forced BDC trend again: 1970-1994 predates most of it. Variance is the cleaner result. Every diagnostic sits inside the window, so CESM's year-to-year spread is stable across the record even where its mean is not. Major SSW: 15 in 1970-1994 against an expected 16.4 [9, 25] at the 0.66/winter anchor rate— pass.